Stata Coding Sample

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This coding sample presents the validation and preferred event-study estimation layer from my senior seminar project on Hurricane Mitch and municipal nighttime lights in Central America. The complete project and replication package are available at github.com/orlando-valladares/mitch-nightlights-replication.

My GitHub also includes replication packages for recent Guatemalan and Honduran drought reports that I have been disseminating to help channel funding into the region in response to the severe 2026 El Niño drought.

The code below runs on the real analysis panel distributed with the repository. It checks the panel structure, applies a pre-treatment sample rule, estimates the 64- and 50-knot two-year event studies with municipality and year fixed effects, clusters inference by municipality, and exports machine-readable results. The omitted period is 1996–1997.

```
version 18
clear all
set more off

* Run from the repository root.
local panel "working/data/processed/analysis_panel_main.csv"
local output "out/results/stata/preferred_wind_event_study.csv"
local ntl_cutoff 101.79

capture confirm file "`panel'"
if _rc {
    display as error "Missing analysis panel: `panel'"
    exit 601
}

import delimited using "`panel'", clear varnames(1)
confirm variable muni_unit_id year log_ntl pre_mean_stable_sum
confirm variable ebtrk_treat50_r64_all ebtrk_treat50_r50_all

* The panel must be balanced and uniquely identified before estimation.
isid muni_unit_id year
assert inrange(year, 1992, 2013)
assert !missing(muni_unit_id, year, log_ntl, pre_mean_stable_sum)
bysort muni_unit_id: assert _N == 22

* This restriction uses only the six pre-treatment years, 1992-1997.
keep if pre_mean_stable_sum >= `ntl_cutoff'
quietly count
assert r(N) == 10956

tempfile estimates
tempname results
postfile `results' str4 wind int period_start period_end double coefficient ///
    std_error p_value ci_low ci_high long observations clusters using `estimates', replace

foreach wind in r64 r50 {
    local treatment ebtrk_treat50_`wind'_all
    bysort muni_unit_id: assert `treatment' == `treatment'[1]

    local event_terms
    foreach period in 1992_1993 1994_1995 1998_1999 2000_2001 2002_2003 ///
        2004_2005 2006_2007 2008_2009 2010_2011 2012_2013 {
        gettoken start end : period, parse("_")
        local end = subinstr("`end'", "_", "", .)
        generate double event_`start'_`end' = ///
            `treatment' * inrange(year, `start', `end')
        local event_terms `event_terms' event_`start'_`end'
    }

    * 1996-1997 is omitted. Municipality and year effects are absorbed;
    * inference is clustered at the municipality level.
    quietly areg log_ntl `event_terms' i.year, ///
        absorb(muni_unit_id) vce(cluster muni_unit_id)

    local residual_df = e(df_r)
    local critical = invttail(`residual_df', 0.025)
    foreach period in 1992_1993 1994_1995 1998_1999 2000_2001 2002_2003 ///
        2004_2005 2006_2007 2008_2009 2010_2011 2012_2013 {
        gettoken start end : period, parse("_")
        local end = subinstr("`end'", "_", "", .)
        local term event_`start'_`end'
        local b = _b[`term']
        local se = _se[`term']
        local p = 2 * ttail(`residual_df', abs(`b' / `se'))
        post `results' ("`wind'") (`start') (`end') (`b') (`se') (`p') ///
            (`b' - `critical' * `se') (`b' + `critical' * `se') ///
            (e(N)) (e(N_clust))
    }
    post `results' ("`wind'") (1996) (1997) (0) (.) (.) (.) (.) ///
        (e(N)) (e(N_clust))
```

```
    drop event_*
}

postclose ‘results’
use ‘estimates’, clear
sort wind period_start
isid wind period_start period_end
export delimited using "‘output’", replace

display as result "Preferred wind event studies exported to ‘output’."
```